

Partial differential equations

$$F(u, \frac{\partial u}{\partial x_1}, \frac{\partial u}{\partial x_2}, \dots, \frac{\partial u}{\partial t}, \frac{\partial^2 u}{\partial x_1^2}, \dots, \frac{\partial^2 u}{\partial x_2 \partial t}, \dots, x_1, \dots, t) = 0$$

PDEs solvable as ODEs:

12.1. / 17) $u(x, y): \frac{\partial^2 u}{\partial x^2} + 16\pi^2 u = 0$

$$u_{xx} + 16\pi^2 u = 0$$

fix $\rightarrow v(x) = u(x, y) \rightarrow$ ODE

$$v(x) = A \cdot \cos(4\pi x) + B \sin(4\pi x)$$

$$\rightarrow u(x, y) = A(y) \cos(4\pi x) + B(y) \sin(4\pi x)$$

19) $u_y + y^2 u = 0$

$v(y) := u(x, y)$ x fixed

$$\frac{dv}{v} = -y^2 dy$$

$$\ln v = -y^3/3 + C'$$

$$v(y) = e^{C'} \cdot e^{-y^3/3} = C \cdot e^{-y^3/3}$$

$$u(x, y) = C(x) \cdot e^{-y^3/3}$$

22) $u_{xy} = u_x$

$v(y) := u_x(x, y) = \frac{\partial u(x, y)}{\partial x} \quad \Rightarrow$ fix x

$$v' = v$$

$$\frac{dv}{v} = dy$$

$$\ln v = y + C'$$

$$v = C \cdot e^y$$

$$u_x(x, y) = C(x) e^y$$

$$u(x, y) = e^y \underbrace{\int C(x) dx}_{C_1(x)} + C_2(y)$$

Wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$c^2 = \frac{T}{\rho}$$

$$x \in [0, L]$$

$$u(0, t) = 0, \quad u(L, t) = 0$$

boundary conditions

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x)$$

initial conditions

let's set $c=1, L=1$.

$$\rightarrow u_{tt} = u_{xx}$$

12.3/5)

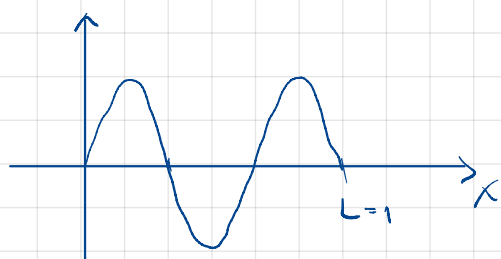
$$f(x) = A \cdot \sin(3\pi x)$$

$f^*(x)$: odd, 2L-periodic extension

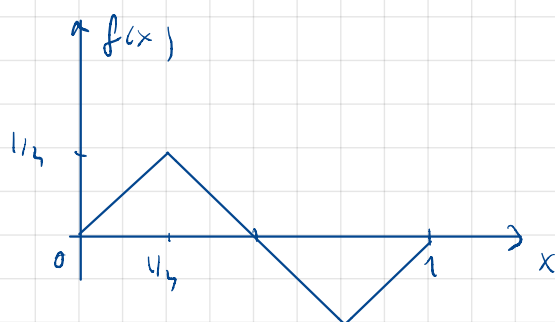
$$\text{D'Alembert: } u(x, t) = \frac{1}{2} [f^*(x-t) + f^*(x+t)]$$

$$\text{Here } f^*(x) = f(x)$$

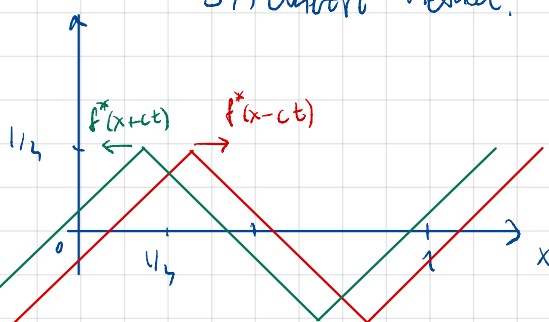
$$\begin{aligned} u(x, t) &= \frac{A}{2} [\sin(3\pi x - 3\pi t) + \sin(3\pi x + 3\pi t)] \\ &= A \sin(3\pi x) \cos(3\pi t) \end{aligned}$$



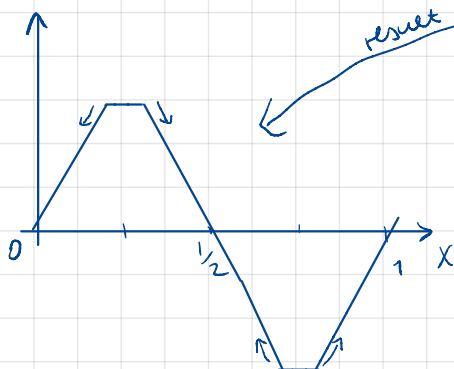
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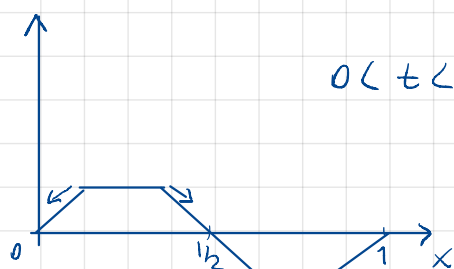
D'Alembert method:



So:



time
→



$$1/4 \leq t \leq 1/2 : u(x, t) = -u(x, 1/2 - t)$$

$$1/2 \leq t \leq 1 : u(x, t) = u(x, 1 - t)$$

$$1/2 \leq t \leq 3/4 : u(x, t) = -u(x, t - 1/2)$$

Then periodic with $T=1$.

Uniform elastic beam (rod)

$$u_{tt} = -c^2 u_{xxxx}$$

$$c^2 = \frac{E \cdot I}{SA}$$

15) $u = F(x) G(t)$

$$F \cdot \ddot{G} = -c^2 F^{(4)} G$$

$$\frac{F^{(4)}}{F} = -\frac{1}{c^2} \frac{\ddot{G}}{G} = \text{const.} =: \beta^4$$

$$F^{(4)} = \beta^4 F$$

↓

$$F = e^{ax} \rightarrow a^4 \cdot e^{ax} = \beta^4 \cdot e^{ax}$$

$$a^4 = \beta^4 \Rightarrow a = \pm \beta, \pm i\beta$$

$$\ddot{G} = -c^2 \beta^4 \cdot G$$

↓

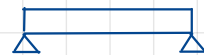
$$G = e^{bx} \rightarrow b^2 e^{bx} = -c^2 \beta^4 e^{bx}$$

$$b = \pm i c \beta$$

$$F = A \cos(\beta x) + B \sin(\beta x) + C \cosh(\beta x) + D \sinh(\beta x)$$

$$G = a \cos(\beta^2 c t) + b \sin(\beta^2 c t)$$

16) Simply supported beam



$$u(0, t) = 0$$

$$A + C = 0$$

$$u(L, t) = 0$$

$$A \cos(\beta L) + B \sin(\beta L) + C \cosh(\beta L) + D \sinh(\beta L) = 0$$

$$u_{xx}(0, t) = 0$$

$$-A\beta^2 + C\beta^2 = 0$$

$$u_{xx}(L, t) = 0$$

$$-A\beta^2 \cos(\beta L) - B\beta^2 \sin(\beta L) + C\beta^2 \cosh(\beta L) + D\beta^2 \sinh(\beta L) = 0$$

$$A = C = 0$$

$$B \sin(\beta L) + D \sinh(\beta L) = 0$$

$$\Rightarrow D = 0$$

$$-B \sin(\beta L) + D \sinh(\beta L) = 0$$

$$\beta L = n \cdot \pi$$

$$u(x,t) = \sum_{n=1}^{\infty} B_n \sin(\beta_n x) \cdot (\cos(\beta_n^2 c t) + b_n \sin(\beta_n^2 c t))$$

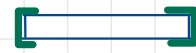
$$\omega_n = c \cdot \beta_n^2 \quad ! \quad / \text{ string: } \omega_n = c \cdot \beta_n \quad /$$

$$17) \quad u(x,0) = f(x) = x(L-x)$$

$$\sum_{n=1}^{\infty} B_n \sin(\beta_n x) = x(L-x)$$

$$\rightarrow B_n = \frac{4L^2}{\pi^3} \frac{1 - (-1)^n}{n^3}$$

$$19) \quad \text{Clamped beam:}$$



$$u(0,t) = 0$$

$$A + C = 0$$

$$u(L,t) = 0 \quad A \cosh(\beta L) + B \sinh(\beta L) + C \cos(\beta L) + D \sin(\beta L) = 0$$

$$u_x(0,t) = 0 \quad \beta(B + D) = 0$$

$$u_x(L,t) = 0 \quad -A\beta \sinh(\beta L) + B\beta \cosh(\beta L) + C\beta \sinh(\beta L) + D\beta \cos(\beta L) = 0$$

$$\left. \begin{aligned} A \cosh(\beta L) + B \sinh(\beta L) - A \cos(\beta L) - B \sin(\beta L) &= 0 \\ -A \sinh(\beta L) + B \cosh(\beta L) - A \sinh(\beta L) - B \cos(\beta L) &= 0 \end{aligned} \right\}$$

$$\begin{pmatrix} \cosh(\beta L) - \cos(\beta L) & \sinh(\beta L) - \sin(\beta L) \\ -\sinh(\beta L) - \cosh(\beta L) & \cosh(\beta L) - \cos(\beta L) \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = 0$$

$$\det: (\cosh(\beta L) - \cos(\beta L))^2 + (\sinh(\beta L) - \sin(\beta L))(\sinh(\beta L) + \sin(\beta L))$$

$$= \cosh^2(\beta L) - 2\cosh(\beta L)\cos(\beta L) + \cos^2(\beta L) + \sinh^2(\beta L) - \sin^2(\beta L) =$$

$$= 1 + 1 - 2\cosh(\beta L)\cos(\beta L) \stackrel{!}{=} 0$$

$$\rightarrow \cosh(\beta L)\cos(\beta L) = 1$$

$$\text{If } \beta L \gg 1 \Rightarrow \cos(\beta L) \approx 0 \Rightarrow \beta \approx \pi/2 + n\pi \quad n \in \mathbb{Z}$$

Wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

- Separating variables

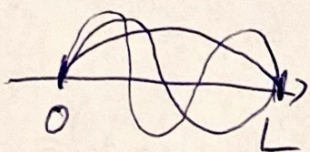
$$u(x, t) = F(x) G(t)$$

$$\frac{1}{c^2} \frac{\ddot{G}}{G} = \frac{F''}{F} = k$$

$$\begin{aligned} F'' &= kF \\ \ddot{G} &= c^2 k G \end{aligned} \quad (k \text{ unknown})$$

- Boundary conditions

$$F(0) = F(L) = 0$$



$$\Rightarrow F(x) = \sum_{n=1}^{\infty} B_n \underbrace{\sin \lambda_n x}_{F_n(x)}$$

$$\lambda_n = \frac{n\pi}{L}$$

$$\Downarrow$$
$$k_n = -\lambda_n^2$$

$$\Downarrow$$
$$\ddot{G}_n = c^2 k_n G_n$$

$$\Downarrow$$
$$G_n = \alpha_n \cos(\lambda_n c t) + \beta_n \sin(\lambda_n c t)$$

$$u(x, t) = \sum_{n=1}^{\infty} (\alpha_n \cos(\lambda_n c t) + \beta_n \sin(\lambda_n c t)) \sin(\lambda_n x)$$

Initial condition

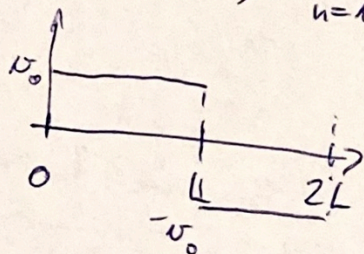
$$\text{Let } u(x, t=0) = 0$$

$$\frac{\partial u}{\partial t}(x, t=0) = v_0$$

From general solution

$$u(x, t=0) = \sum_{n=1}^{\infty} \alpha_n \sin(\lambda_n x) = 0 \Rightarrow \alpha_n = 0$$

$$u_t(x, t=0) = \sum_{n=1}^{\infty} c \beta_n \lambda_n \sin(\lambda_n x) = v_0$$



$c \beta_n \lambda_n$ are Fourier coefficients of square wave

$$\begin{aligned} c \beta_n \lambda_n &= \frac{1}{L} \int_0^{2L} f(x) \sin(\lambda_n x) dx = \frac{2}{L} \int_0^L v_0 \sin(\lambda_n x) dx = \\ &= \frac{2v_0}{L} \left[-\frac{1}{\lambda_n} \cos(\lambda_n x) \right]_0^L = -\frac{2v_0}{n\pi} ((-1)^n - 1) = \begin{cases} 0 & n \text{ even} \\ \frac{4v_0}{n\pi} & n \text{ odd} \end{cases} \end{aligned}$$

$$\beta_n = \frac{1}{c \lambda_n} \frac{2v_0}{n\pi} (1 - (-1)^n) = \frac{2L v_0}{c(n\pi)^2} (1 - (-1)^n)$$

$$u(x, t) = \sum_{n=1}^{\infty} \beta_n \sin(\lambda_n x) \cdot \sin(\lambda_n c t)$$

Heat equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Separation of variables

$$\frac{1}{c^2} \frac{\dot{G}}{G} = \frac{F''}{F} = -p^2$$

$$F'' = -p^2 F$$

$$\dot{G} = -c^2 p^2 G$$

1., Fixed temperature BC $F(0) = F(L) = 0$

$$\Rightarrow F_n(x) = \sin \lambda_n x$$

$$\lambda_n = \frac{n\pi}{L}$$

2., Insulated BC (Neumann) $\frac{\partial F}{\partial x}(0) = \frac{\partial F}{\partial x}(L) = 0$

$$\Rightarrow F_n(x) = \cos \lambda_n x$$

$$\lambda_n = \frac{n\pi}{L}$$

$$p^2 = \lambda_n^2 \Rightarrow G_n(t) = \alpha_n e^{-\lambda_n^2 c^2 t}$$

General solution

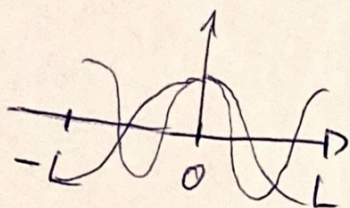
$$u(x,t) = \sum_n F_n(x) \alpha_n e^{-\lambda_n^2 c^2 t}$$

Initial condition $u(x, t=0) = u_0(x)$

Neumann BC $\frac{\partial u}{\partial x}(x=0, t) = \frac{\partial u}{\partial x}(x=L, t) = 0$

From general solution

$$u(x, t=0) = \sum_{n=0}^{\infty} \alpha_n \cos(\lambda_n x) = u_0(x)$$

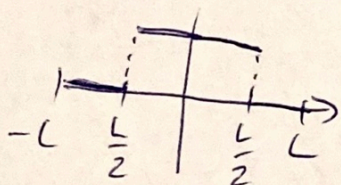


Extend u_0 as even function
 $\Rightarrow$ Fourier coefficients

$$\alpha_n = \frac{2}{L} \int_0^L u_0(x) \cos(\lambda_n x) dx$$

Example

$$u_0 = \begin{cases} 1 & 0 < x < L/2 \\ 0 & L/2 < x < L \end{cases}$$



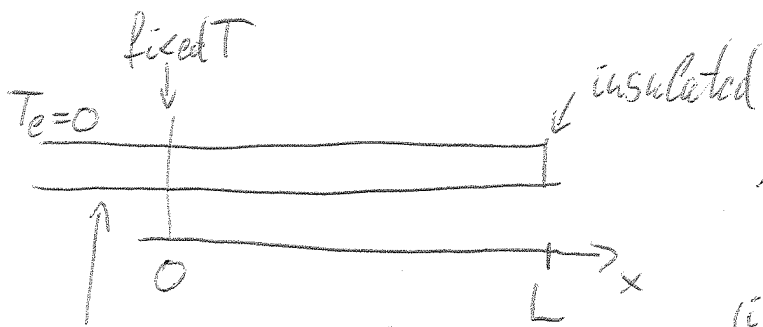
$$\begin{aligned} \alpha_n &= \frac{2}{L} \int_0^{L/2} \cos(\lambda_n x) dx = \frac{2}{L} \frac{1}{\lambda_n} \left[\sin(\lambda_n x) \right]_0^{L/2} \\ &= \frac{2}{n\pi} \left(\sin\left(\frac{n\pi}{2}\right) \right) = \frac{2}{n\pi} \begin{cases} 0 & \text{if } n \equiv 0 \pmod{2} \\ 1 & \text{if } n \equiv 1 \pmod{4} \\ -1 & \text{if } n \equiv 3 \pmod{4} \end{cases} \end{aligned}$$

$$\alpha_0 = \frac{1}{L} \int_0^{L/2} 1 dx = \frac{1}{2}$$

$$u(x, t) = \sum_{n=0}^{\infty} \alpha_n \cos(\lambda_n x) e^{-c^2 \lambda_n^2 t}$$

$$u(x, t \rightarrow \infty) = \alpha_0 \quad \text{average}$$

Heat conduction with mixed BC



in good heat conducting
water bath
→ fixed temperature

Boundary conditions:

(i) $u(0, t) = T_e = 0$ Dirichlet

(ii) $u_x(L, t) = 0$ Neumann

Initial condition:

$u(x, 0) = T_0$ uniform T

Separation of variables → $u(x, t) = F(x) G(t)$

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$F'' = -\rho^2 F$$

$$G' = -c^2 \rho^2 G$$

Solution for F: $F(x) = A \sin(\lambda x) + B \cos(\lambda x)$

(i) $F(0) = 0 \Rightarrow B = 0$

(ii) $\frac{\partial F}{\partial x}(L) = 0 \Rightarrow \lambda \cos(\lambda L) = 0 \Rightarrow \lambda L = \frac{\pi}{2} + n\pi$
 $\Rightarrow \lambda_n = \frac{\pi}{2L} (2n+1)$

→ Solutions for F with BC:

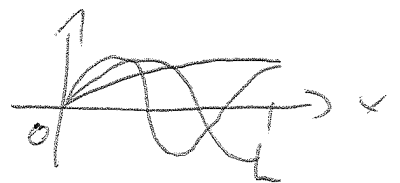
$$F_n(x) = \sin\left(\frac{\pi}{2L} (2n+1)x\right)$$

$$F''_n = -\lambda_n^2 F \Rightarrow \rho = \lambda_n$$

$$G'_n = -c^2 \lambda_n^2 G \Rightarrow G_n(t) = e^{-\lambda_n^2 c^2 t}$$

→ General solution:

$$u(x, t) = \sum_{n=0}^{\infty} \underbrace{\alpha_n}_{\text{constants}} F_n(x) G_n(t) = \sum_{n=0}^{\infty} \alpha_n \sin\left(\frac{\pi}{2L} (2n+1)x\right) e^{-\lambda_n^2 c^2 t}$$



• Initial condition

$$u(x, 0) = \sum_n \alpha_n \sin\left(\frac{\pi}{2L} (2n+1)x\right) = T_0 \quad \text{for } 0 < x < L$$

looks like a Fourier series of $4L$ periodic function

$$f(x) = \sum_n \left[a_n \cos\left(\frac{2\pi n}{4L} x\right) + b_n \sin\left(\frac{2\pi n}{4L} x\right) \right]$$

but $a_n = 0$ and $b_n = 0$ if n even.

extend $u(x, 0)$ to $4L$ period

odd: $a_n = 0$

same in $[0, L]$ and $[L, 2L]$

$\Rightarrow b_n = 0$ for n even

$$\alpha_n = b_{2n+1}$$

Compute Fourier series of extended function

$$b_n = \frac{1}{2L} \int_{-2L}^{2L} \text{sign}(x) \sin\left(\frac{2\pi n}{4L} x\right) dx = \frac{T_0}{L} \int_0^{2L} \sin\left(\frac{\pi n}{2L} x\right) dx =$$

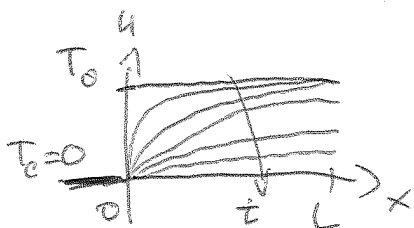
$$= \frac{T_0}{L} \frac{2L}{\pi n} \left[-\cos\left(\frac{\pi n}{2L} x\right) \right]_0^{2L} = -\frac{2T_0}{\pi n} \left(\underbrace{-\cos(\pi n)}_{(-1)^n} - 1 \right) = \begin{cases} \frac{4T_0}{\pi n} & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

$$\alpha_n = b_{2n+1} = \frac{4T_0}{\pi(2n+1)}$$

Solution of initial value problem:

$$u(x, t) = \sum_{n=0}^{\infty} \frac{4T_0}{\pi(2n+1)} \sin\left(\frac{\pi}{2L} (2n+1)x\right) e^{-\left(\frac{\pi}{2L} (2n+1)\right)^2 c^2 t}$$

higher n decays faster



Note: if $T_0 \neq 0$ solution $u(x, t) + T_0$